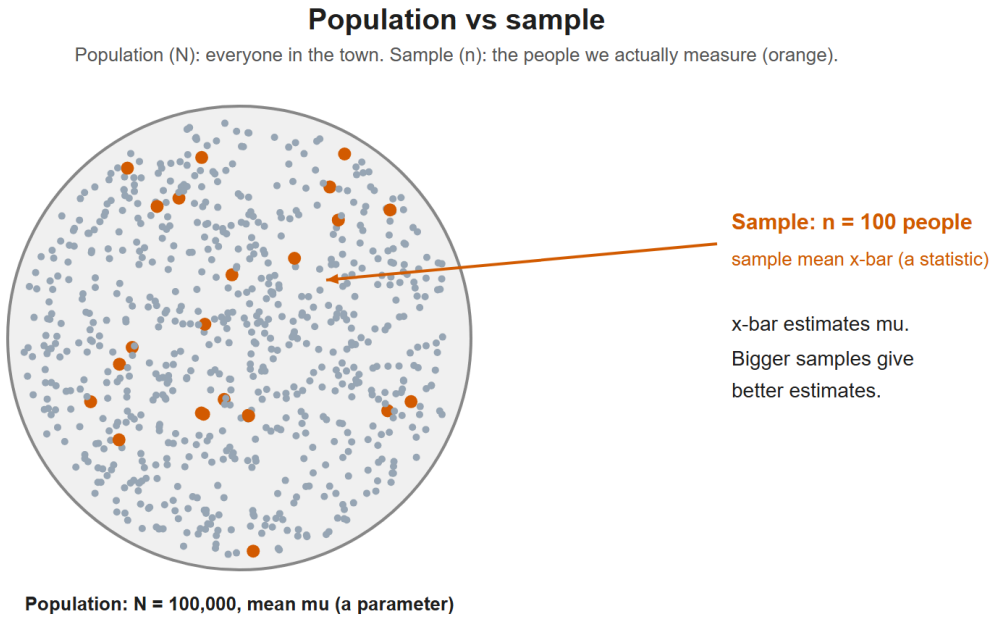
Lesson 3.9 - Sample Mean and Population Mean

> **Video:** Sample Mean and Population Mean by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Population vs sample

- **Population** (size N): everyone of interest. Example: all **100,000** people in city XYZ.
- **Sample** (size n): the part we actually measure. Example: **100** people chosen from the city.

Weighing all 100,000 residents is impractical, so we weigh a sample of 100 and use its average to **estimate** the average weight of the whole city.



The population is everyone (grey dots); the sample is the small group actually measured (orange). The sample mean x-bar is used to estimate the population mean mu.

Formulas

Sample mean (a **statistic**):

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Population mean (a **parameter**):

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N}$$

The formulas look the same. The only difference is **lowercase** n (sample size) versus **capital** N (population size).

	Symbol	Size	Called a
Sample mean	$\bar{x}$	n	**statistic**
Population mean	μ	N	**parameter**

Bigger samples give better estimates

We want $\bar{x}$ to be as close as possible to μ. **Increasing the sample size** makes $\bar{x}$ a better estimate of μ. Going from $n = 100$ to $n = 1{,}000$ gives a more accurate estimate, but it is ten times as much work.

> **Trade-off:** more accuracy requires more data, and more data requires more effort.

Key takeaways

- $\bar{x}$ (sample, n) is a **statistic**. μ (population, N) is a **parameter**.
- Both are "sum ÷ count". Only the group being averaged differs.
- Larger samples make $\bar{x}$ a more reliable estimate of μ.